98/100

CHE 313 Transport Phenomena III

Exam #1 (2/19/2020)

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We are separating acetone and ethanol in a distillation column with a total condenser and a partial reboiler. The column has two feeds and operates at a pressure of 1.0 atm. Feed 1 is a saturated liquid that is 60.0 mol% acetone and has a feed rate of F1 = 200.0 kmol/hr. The second feed is 40.0 mol% acetone, has a feed rate of F2 = 300.0 kmol/hr, and is a two-phase mixture that is 20% vapor. Assume CMO is valid. Bottoms product is 10.0 mol% acetone and distillate product is 80.0 mol% acetone. External reflux ratio L/D = 2.0.

1) (5 pts) Draw a schematic diagram of the distillation column and label all known variables (use V & L for enriching section, V' & L' for middle section, and $\bar{L}$ & $\bar{V}$ for stripping section)
2) (10 pts) Draw mass balance envelopes for each section (total four balance envelopes)
3) (15 pts) Find D, B, L, V, V', L', $\bar{L}$, and $\bar{V}$ (hint: set up overall mass balance and component mass balance)
4) (5 pts) Find slope, y-intercept, and coordinate on y=x curve of the top operating line:

$$y = \left(\frac{L}{V}\right)x + \left(1 - \frac{L}{V}\right)x_D$$

5) (5 pts) Find slope, x-intercept, and coordinate on y=x curve of the bottom operating line:

$$y = \left(\frac{\bar{L}}{\bar{V}}\right)x - \left(\frac{\bar{L}}{\bar{V}} - 1\right)x_B$$

6) (10 pts) The equation for the feed line is:

$$y = \frac{q_i}{q_i - 1}x + \frac{z_i}{1 - q_i}$$

$$q_i = \frac{L_{Fi}}{F_i}$$

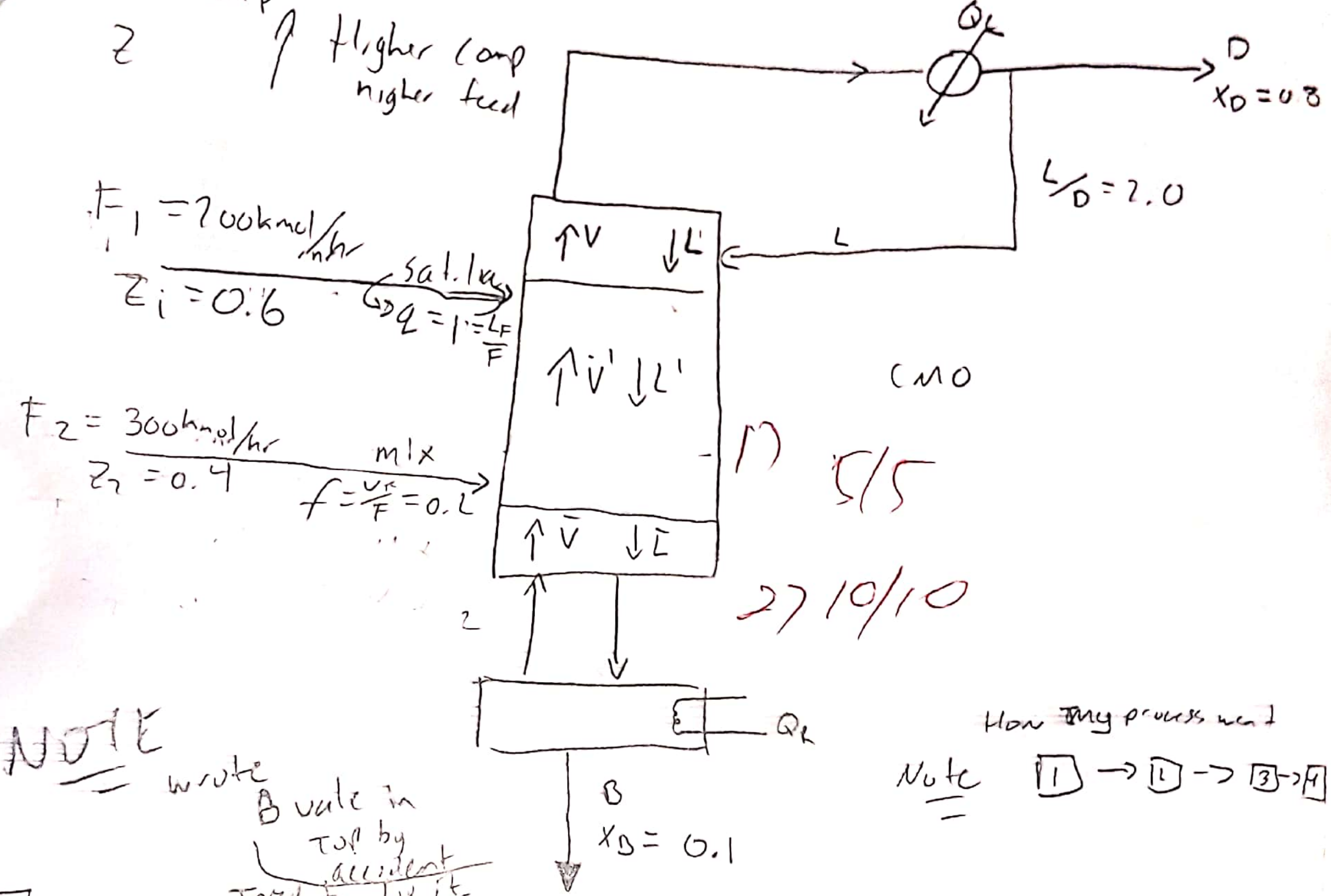
Find q_1, q_2, coordinates on y=x line, and slopes of two feed lines (F1 & F2), and x-intercept of F2.

7) (5 pts) Write overall mass balance and component mass balance around the top column (hint: mass balance between top feed and distillate)
8) (15 pts) Derive an operating equation for the middle section. Find slope and y-intercept
9) (10 pts) Draw the top operating, bottom operating, two feeds, and middle operating lines

Ethanol

Z comp
Z ↑ Higher comp higher feed

David Abarado



CMO

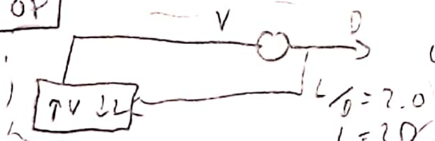
1) 5/5

27 10/10

NOTE
wrote B vale in Top by accident
Tried to fix it

How They process went
Note $\boxed{1} \to \boxed{2} \to \boxed{3} \to \boxed{4}$

2) $\boxed{2}$ TOP



$V = D + L$

3)

In = Out

OMB: $V = L + D \Rightarrow D = V - L$

MVCB: $Vy = Lx + Dx_D$

$D = V - 2D$

$3D = V$

$L = 2D$

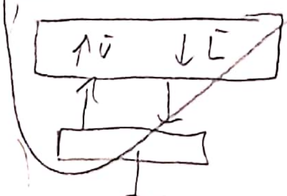
$L = 2(271.426)$

$\boxed{L = 542.856 \text{ kmol/hr}}$

~~$V = 3(271.4287)$~~

$\boxed{V = 814.284 \text{ kmol/hr}}$

Bottom

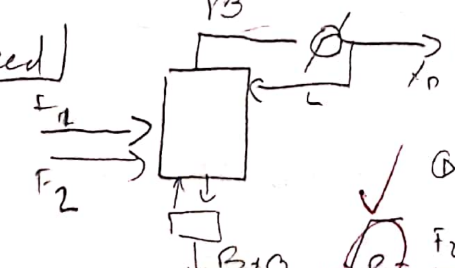


MB: $\bar{L} = \bar{V} + B$

MVCB: $\bar{L}x = \bar{V}y + Bx_B$

3) 15/15

$\boxed{1}$ Feed



MB: ① $F_2 + F_1 = D + B \Rightarrow D = F_2 + F_1 - B$

MVCB: ② $F_2 z_2 + F_1 z_1 = Dx_D + Bx_B$

① → ② $\Rightarrow F_2 z_2 + F_1 z_1 = (F_2 + F_1 - B)x_D + Bx_B$

$F_2 z_2 + F_1 z_1 - (F_2 + F_1)x_D = B(x_B - x_D)$

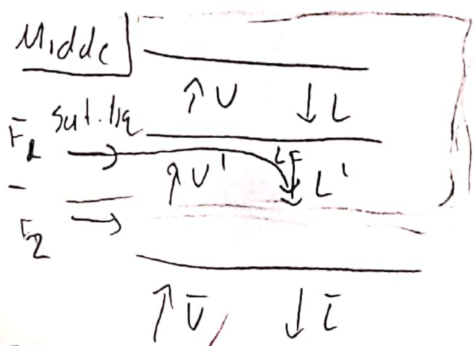
$B = \frac{F_2 z_2 + F_1 z_1 - (F_2 + F_1)x_D}{x_B - x_D} = \frac{(300)(0.4) + (200)(0.6) - (300 + 200)0.8}{\ldots} = 228.57 \text{ kmol/hr}$

D2)

$D = F_2 + F_1 - B$
$= (200) + (300) - 228.571$
$\boxed{D = 271.420 \text{ kmol/h}}$

3) F_1

Middle



Vapor: $\boxed{V = V' = 814.284 \text{ kmol/hr}}$

Liquid: $L + \underbrace{F_1}_{\text{pure liquid}} = L'$

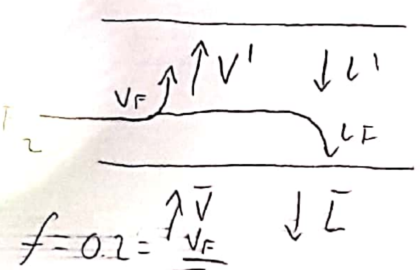
$F_1 = L_F$

Sat. vap. $q = 1 = \frac{L_F}{F}$

$L' = (542.856) + (200)$

$\boxed{L' = 742.8564 \text{ kmol/hr}}$

4) Middle F_2



Vapor: $V_F + \bar{V} = V'$

$\bar{V} = V' - V_F$
$= V' - (0.2F_2)$
$= (814.284) - (0.2)(300)$
$\boxed{\bar{V} = 754.284 \text{ kmol/hr}}$

Liquid: $L_F + L' = \bar{L}$

$\bar{L} = (0.8F_2) + L'$
$= 0.8(300) + 742.8564$
$\boxed{\bar{L} = 982.856 \text{ kmol/hr}}$

$f = 0.2 = \frac{V_F}{F}$

$0.2F_2 = V_F$

$f + q = 1$

$q = 1 - f = 1 - 0.2 = 0.8$

$q = \frac{L_F}{F_2}$

$0.8F_2 = L_F$

4) TOP

$y = \frac{L}{V}x + \left(1 - \frac{L}{V}\right)x_D$

Slope: $\frac{L}{V} = \frac{542.856}{814.284} = \frac{2}{3}$

y int: $\left(1 - \frac{L}{V}\right)x_D = \left(1 - \frac{2}{3}\right)0.8 = \frac{4}{15}$

coord: $y = x = x_D = 0.8$

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5) Bottom

$y = \left(\frac{\bar{L}}{\bar{V}}\right)x - \left(\frac{\bar{L}}{\bar{V}} - 1\right)x_B$

Slope: $\frac{\bar{L}}{\bar{V}} = \frac{982.856}{754.284} = 1.303$

y int: $-\left(\frac{\bar{L}}{\bar{V}} - 1\right)x_B = -(1.3031 - 1)(0.1) = -0.0303$

3/5

X-intercept: $x = 0.0232$

coord: $(0.1, 0.1)$ on $y = x$

from graph w/ $x = 0.65 \Rightarrow y = 0.87$ X

Scanned with CamScanner

$$\frac{q_i}{q_i-1}x + \frac{z_i}{1-q_i} \qquad q_i = \frac{L_{Fi}}{F_i}$$

$\boxed{1}$ Sat. Liq

$$q_1 = 1 = \frac{L_{F_1}}{F_1}$$

$$\text{-Slope} = \frac{q_1}{q_1-1} = \frac{1}{1-1} = \frac{1}{0} = \infty$$

Coord: $y = x = z_1 = \underline{0.6}$

$\boxed{F_2}$ $q_2 + f_2 = 1 \Rightarrow q_1 = 1 - f_2 = 1 - 0.2 = \underline{0.8}$

$$\text{Slope}: \frac{q_2}{q_2-1} = \frac{0.8}{0.8-1} = \underline{-4}$$

10/10

Coord: $y = x = z_2 = \underline{0.4}$

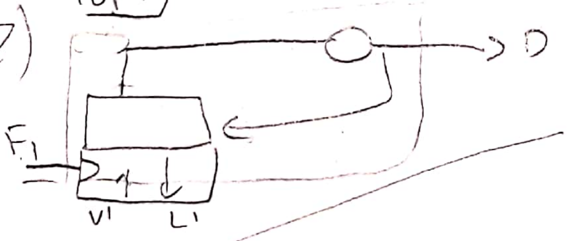
Xintc: $y = 0 \Rightarrow 0 = -4x + \frac{(0.4)}{(1-0.8)}$

$$0 = -4x + 2$$

$$-2 = -4x$$

$$x = \underline{1/2}$$

7) Top +)



MB: $\underline{F_1 + V' = D + L'}$

MVCB: $\underline{F_1 z_1 + V'y = Dx_D + L'x}$

5/5

8) $V'y = L'x + Dx_D - F_1 z_1$

$$\left[y = \frac{L'}{V'}x + \frac{Dx_D - F_1 z_1}{V'}\right]$$

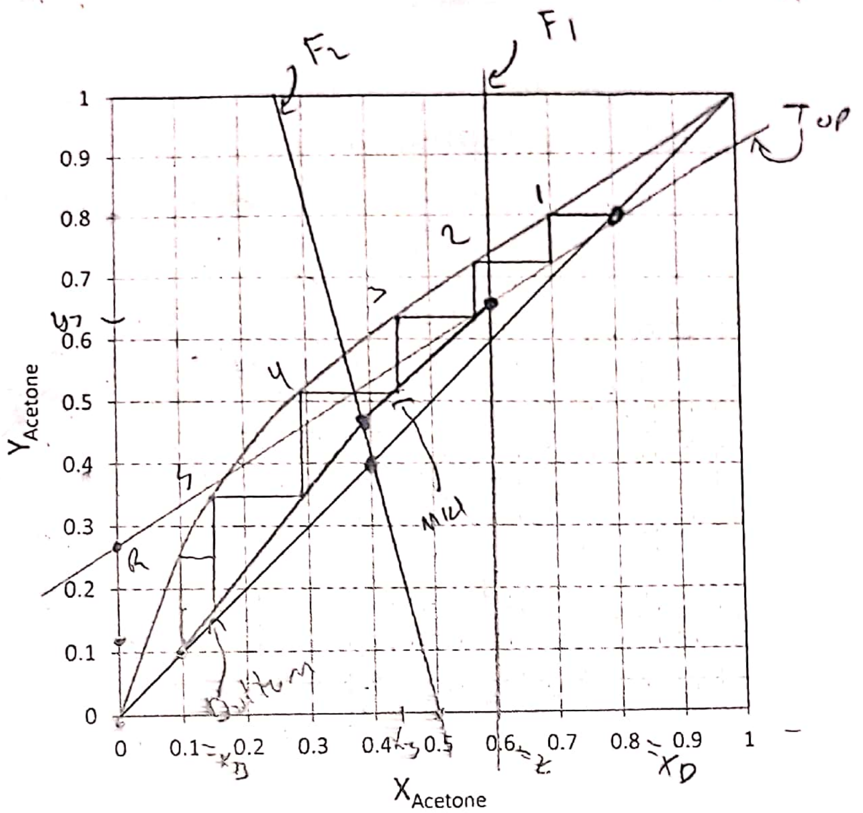
$$\text{Slope}: \frac{L'}{V'} = \frac{742.856}{814.284} \; 0.91228$$

$$\text{y-Intc}: \frac{Dx_D - F_1 z_1}{V'} = \frac{(271.498)(0.8) - (200)(0.6)}{814.284} = 0.114$$

15/15

9)

10/10



10/10

10) (10 pts) Find the number of stages and the optimum locations of feed 1 and feed 2 stages

11) (10 pts) Find mole fraction (x_3, y_3) of the stage 3

10/10

$F_1 = 2$

$F_2 = 4$

= 5

11)

$x_3 = 0.42$

$y_3 = 0.62$

Cool